

A Comparison Study of Infrastructure-Independent Communication Approaches for Rural Emergency Messaging

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Abstract—Communication during a flood, fire, medical incident, power outage, or cellular-network failure is difficult when users are distributed across rural or peri-rural areas. This paper reviews and compares communication approaches that may support short emergency messages without assuming continuous Internet access. The study contrasts conventional cellular and Internet messaging, Wi-Fi/Bluetooth ad-hoc links, LoRaWAN, general-purpose LoRa mesh systems, and the proposed LoRaResQ workflow. The comparison uses infrastructure dependence, network topology, payload suitability, range and energy characteristics, deployment complexity, and operational and security support as evaluation criteria. The literature indicates that LoRa is attractive for small, delay-tolerant messages because of its long range and low power requirements, but also has important limitations: low throughput, shared-channel contention, duty-cycle constraints, and variable propagation. Existing projects demonstrate the feasibility of low-power LoRa communication, but leave a practical gap between a flexible technical mesh and a simple, alert-first rural operating workflow. LoRaResQ addresses that gap as a proposed, low-cost system that connects a phone to a personal LoRa node and allows eligible packets to be relayed by nearby participant nodes. The paper does not claim hardware validation; the proposed evaluation plan identifies the evidence required before deployment or safety-critical use.

Index Terms—LoRa, LoRaWAN, mesh networking, emergency communication, disaster resilience, rural connectivity, off-grid communication.

I. INTRODUCTION

Mobile phones are powerful emergency tools only when a working device, electricity supply, radio link, cellular tower, backhaul, and subscribed service are available. Rural roads, farms, forest edges, shelters, and disaster-affected localities may not satisfy all of these conditions. A communication fallback does not need to carry video or general Internet traffic to be useful. It may only need to communicate a bounded message such as “medical help needed”, “road blocked”, “I am safe”, or “assemble at the school”.

LoRa is a long-range, low-power physical-layer technology designed for small packets. LoRaWAN is a network protocol and system architecture built around star-of-stars gateways, while a LoRa mesh uses peer nodes that can forward packets. These approaches should not be treated as interchangeable. Adelantado et al. show that LoRaWAN performance is constrained by capacity, collisions, and scalability assumptions [1]. Augustin et al. similarly describe the range, data-rate, and energy trade-offs of LoRa-based networks [2]. These limitations

matter more in an emergency because an alert should be useful without creating the false impression of guaranteed delivery.

This paper presents a focused comparison for the design of LoRaResQ, a proposed community-owned emergency messaging mesh. The contribution is not a new radio modulation or a claim of superiority over existing platforms. Instead, it is a requirements-led comparison and a clearly bounded system proposal that can be implemented and tested with a small number of nodes.

II. REVIEW METHOD

The study searched Google Scholar, publisher pages, arXiv, and official standards pages in October 2026 using combinations of “LoRa”, “LoRaWAN”, “mesh”, “off-grid”, “emergency communication”, “disaster”, “rural”, “scalability”, and “Meshtastic”. Sources were included when they were journal articles, scholarly preprints, or primary technical specifications with a clear description of architecture, constraints, or measured results. The six scholarly papers selected for this focused comparison are mostly adjacent LoRa, LoRaWAN, mesh, and Meshtastic studies; LoRaMoto is an emergency-specific simulation study rather than a field trial. This limitation is recorded explicitly below. Sources were also screened for direct relevance to short-message communication rather than general high-bandwidth networking. Official technical documentation is used only to clarify terminology, regulation, and protocol behaviour; it is not treated as independent experimental evidence. Conference proceedings were not included in this compact review corpus so that the comparison would use sources with complete, inspectable methods and conclusions; this choice may exclude relevant emergency-network results.

The comparison criteria are:

- 1) **Infrastructure dependence:** whether the system requires a cellular network, Internet, gateway, or cloud service.
- 2) **Network topology:** direct, star, gateway-based, or multi-hop peer communication.
- 3) **Payload suitability:** whether the system is appropriate for short text and structured alerts rather than high-bandwidth content.
- 4) **Range and energy:** reported or expected suitability for battery-powered portable nodes, without assuming a universal range.
- 5) **Deployment complexity:** hardware, gateway, configuration, and operational requirements.

- 6) **Operational and security support:** delivery states, prioritisation, expiry, localisation, authentication, and privacy.

III. TECHNOLOGIES AND RELATED SYSTEMS

A. Conventional connected communication

Cellular data, SMS, and Internet messaging provide familiar user experiences and can support large payloads. Their weakness in this context is dependence on external infrastructure. A damaged tower, congested cell, unavailable backhaul, depleted battery, or missing coverage can prevent a message from being sent. These services should therefore remain the preferred channel when available, while an off-grid system should be treated as a fallback rather than a replacement for official emergency services. This paper treats cellular and Internet messaging as the baseline against which the infrastructure dependence of the proposed system is compared; it does not claim that the baseline is technically inferior.

B. Wi-Fi and Bluetooth ad-hoc links

Wi-Fi and Bluetooth Low Energy are useful for connecting a phone to a nearby device or for local peer-to-peer communication. Bluetooth Low Energy is specified for low-power short-range links [3]; Wi-Fi provides higher throughput than LoRa under the IEEE 802.11 family of local-area network standards [4], but Wi-Fi and Bluetooth links have shorter practical range and greater sensitivity to obstructions and user density. A hybrid architecture can use Bluetooth for the phone-to-node link and LoRa for the outdoor node-to-node link. This separates the user interface from the low-bandwidth radio network.

C. LoRaWAN

LoRaWAN provides a standardised low-power wide-area architecture, normally using end devices, gateways, and network/application servers. Its strengths include mature device classes, long-range links, and low energy consumption. Its architecture is not inherently an infrastructure-free peer mesh: a gateway and network service are normally required to carry data to an application. Recent survey work on scalable LoRaWAN similarly identifies traffic, interference, resource allocation, and topology as important constraints [5]; the limits identified by Adelantado et al. include collisions, data-rate trade-offs, and duty-cycle-related constraints [1]. The LoRaWAN specification defines the end-device, gateway, and network-server architecture used by this comparison [6].

D. LoRa mesh systems

Peer-to-peer LoRa systems remove the requirement for a permanent gateway by allowing nodes to relay selected packets. A recent journal review comparing open-source and proprietary LoRa mesh technologies identifies cost, energy efficiency, latency, range, security, and routing as the main design trade-offs [7]. A mesh can extend reach when a direct path is blocked, but every relay consumes airtime and energy. Duplicate suppression, hop limits, queue policy, packet expiry, and acknowledgement semantics are therefore essential.

LoRaMoto is a closer emergency-specific example: it extends LoRaWAN with end-node forwarding for civilian safety-status messages after an earthquake, but its evaluation is simulation-based rather than a field validation [8].

E. Meshtastic as a practical reference

Meshtastic is an important open-source reference for practical off-grid LoRa mesh communication. Its official documentation describes managed flooding with hop-limited, SNR-aware re-broadcast, next-hop routing for direct messages since firmware 2.6, and CSMA/CA channel access [9]. It demonstrates that inexpensive nodes, user devices, and mesh forwarding can be combined in a community project. Its acknowledgement semantics are also instructive: a broadcast is treated as implicitly acknowledged once the sender hears any node rebroadcast it, whether or not that node holds the channel key [9]. This is the local-acceptance versus recipient-acknowledgement distinction that LoRaResQ's status model is meant to expose. A recent Meshtastic-based campus system reports a maximum field link of approximately 2.47 km under its stated campus conditions [10]. The paper's body does not establish that distance as a statistically validated packet-delivery result: its reported tracker frames, duty-cycle setting, and 22 dBm test configuration require careful interpretation. The authors identify packet-delivery-rate and latency campaigns as future work. This arXiv preprint concerns campus telemetry, not emergency response, and must not be presented as a LoRaResQ range guarantee. LoRaResQ does not claim to replace or outperform Meshtastic. It narrows the user experience to a rural emergency workflow: a small set of preset alerts, bounded text, explicit status, local-language labels, and a deployment plan that can be demonstrated with three interoperable personal nodes.

IV. COMPARISON

Table I compares the alternatives against the requirements of a local emergency-message fallback. The qualitative entries describe architectural fit, not measured performance.

The comparison produces three design conclusions. First, no approach removes the need to state operating conditions: "sent to the local node" is not the same as "received by the intended person". Second, LoRa's advantage is appropriate only when the payload is small and delay is acceptable. Third, the user workflow is a major part of emergency-system reliability. A system that has a radio link but gives ambiguous status, stale alerts, or difficult configuration may fail operationally even if its physical layer works.

V. PROPOSED LORARESQ SYSTEM

LoRaResQ is designed as a small, community-owned mesh for hamlets, farms, forest-edge routes, shelters, or a contained event. Each participant carries a personal node. The phone communicates with its node over Bluetooth LE; the node sends a bounded LoRa packet. An eligible new packet may be forwarded by another participant's node, subject to a hop limit, duplicate cache, queue policy, and packet expiry. The prototype target is three nodes in the path $A \rightarrow B \rightarrow C$, where A and C cannot communicate directly.

TABLE I
QUALITATIVE COMPARISON FOR RURAL EMERGENCY MESSAGING

Approach	Infrastructure	Topology	Payload	Range / energy	Deployment, operations, and security
Cellular / Internet	Required	Provider infrastructure	High bandwidth	Phone-dependent	Mature; unavailable during outage
Wi-Fi ad hoc	Not necessarily	Direct / configurable mesh	High bandwidth, short range	Short range; higher power draw	Configuration-heavy; security depends on setup
Bluetooth LE	Not required	Direct / local relay	Small payloads	Good for short links	Good phone link; not wide-area
LoRaWAN	Gateway and server normally required	Star-of-stars	Small, delay-tolerant	Good low-power range	Mature, but gateway and server required
Generic LoRa mesh	No permanent gateway required	Multi-hop peer	Small, delay-tolerant	Good, relay-dependent	Routing and security policy required
Meshtastic reference	Configurable; can be off-grid	Multi-hop peer	Small text and data	Good, topology-dependent	Mature reference; broad configuration surface; encrypted channels available
LoRaResQ (proposed)	No permanent gateway for prototype	Bounded multi-hop	Alerts and short text	Battery-oriented (target)	Alert-first; status model and security planned; unvalidated, no field data

The application is planned to support:

- structured alerts such as medical help, fire, flood, road blocked, assemble, and all safe;
- bounded free-text messages;
- states that distinguish queued, accepted by the node, transmitted, relayed, and acknowledged;
- local-language labels and icon-based actions;
- priority and expiry rules to prevent stale or excessive traffic; and
- a deployment record containing region, radio configuration, node identity, and test observations.

Security is part of the proposed design rather than an achieved result. For one-to-one messages, the nodes should use authenticated encryption with per-device or per-recipient keys; message identifiers, nonces, hop limits, and replay protection should prevent duplicated or forged packets. Key enrolment and revocation must be defined before field use, and emergency alerts should not contain unnecessary personal or medical data. The prototype will demonstrate the interface and packet-state model first; cryptographic interoperability and resistance to impersonation remain validation tasks. Broadcast alerts need separate protection: because any node can originate a broadcast, alerts should carry sender authentication, for example a per-community key or per-node signature, and relaying nodes should forward one-to-one messages as ciphertext they cannot read.

The system is deliberately not specified as a guaranteed life-safety service. It cannot dispatch an ambulance automatically, guarantee delivery, replace 112, provide Internet access, or substitute for trained responders. When cellular or official emergency communication works, users should use it.

VI. INDIA-SPECIFIC REGULATORY CONTEXT

For a rural Indian prototype, the radio configuration must be selected only after checking the current Wireless Planning and Coordination (WPC) rules. The 2021 exemption rules (G.S.R. 853(E), 10 December 2021) allow low-power short-range devices in the 865–868 MHz band on a non-interference,

non-protection, shared and non-exclusive basis, subject to per-category limits [11]. Two categories are candidates for an alert-messaging node: non-specific short-range devices (25 mW e.r.p., 1% duty cycle, with an FHSS condition on occupied bandwidth) and tracking, tracing and data-acquisition devices (500 mW e.r.p., adaptive power control, duty cycle $\leq 10\%$ for network access points and $\leq 2.5\%$ otherwise, bandwidth ≤ 200 kHz). Which category covers a 125 kHz LoRa emergency-messaging node has not been confirmed with WPC, so the prototype is designed against the stricter limits. Rule 5 requires covered equipment to conform to these limits, to be type approved, and to comply with the corresponding EN standard [11]. An EU868 preset must not be assumed legal in India. The transmit power, bandwidth, and duty-cycle settings used in any trial belong in the deployment record; they are not results claimed by this literature review. The 22 dBm campus configuration reported in the Meshtastic study must not be copied into an India trial: it exceeds the nominal 25 mW limit before antenna-gain and e.r.p. calculations. Firmware should enforce the selected power cap rather than relying on a regional preset.

VII. PROPOSED EVALUATION PLAN

Because the project currently has no hardware, all hardware-related values in this paper are requirements or targets rather than results. The planned validation should include:

- 1) a controlled two-node test with one hundred short messages, selected to provide an initial estimate of delivery proportion rather than a statistically generalisable reliability claim;
- 2) twenty three-node forwarding attempts as a feasibility smoke test, with the direct A-C link unavailable or intentionally attenuated;
- 3) outdoor trials in open and obstructed terrain, recording distance, antenna, spreading-factor settings, transmit power, RSSI/SNR, airtime and duty cycle, packet loss, and acknowledgements;
- 4) battery measurements under a declared workload;
- 5) duplicate and stale-packet tests; and
- 6) phone-to-node Bluetooth Low Energy range, reconnection, and message transfer tests for the hybrid interface;

- 7) a short usability test measuring whether a first-time user can send an alert within thirty seconds without confusing local acceptance with recipient acknowledgement. The thirty-second target is an engineering usability target for the prototype, not a safety standard.

The results should be reported with the test environment and limitations, including delivery proportions with a 95% confidence interval. The two-node and three-node tests are prototype targets; if three radio modules are not available, the forwarding test must be reported as emulated rather than as a three-node hardware result. Vendor range claims should not be presented as project results. No deployment should be described as public-safety-ready until radio compliance, electrical safety, security, and field reliability have been reviewed.

VIII. LIMITATIONS OF THE STUDY

This is a focused design-oriented literature review, not a systematic review with a registered search protocol or meta-analysis. Published systems use different hardware, frequencies, antennas, firmware, terrain, payload sizes, and traffic loads, so their numerical results cannot be transferred directly to LoRaResQ. The proposed system has not yet been tested because the team does not currently possess the radio hardware. The comparison therefore supports the design rationale and test plan; it does not prove range, battery life, reliability, or emergency effectiveness. In addition, the scholarly studies are mostly adjacent technology sources rather than emergency or rural field trials; LoRaMoto is emergency-specific but simulation-based. Excluding conference proceedings may omit relevant disaster communication evidence. The proposed security controls are design requirements, not a completed security evaluation.

IX. CONCLUSION

Cellular and Internet services remain the most capable communication options when available, but they are not dependable as the only channel in every rural or disaster scenario. Wi-Fi and Bluetooth are useful for local links, while LoRa provides a promising low-power long-range link for small messages. LoRaWAN offers standardisation and mature ecosystem support but normally depends on gateway infrastructure. Peer LoRa mesh systems reduce that dependency at the cost of airtime, queueing, forwarding, and operational complexity.

The literature supports the feasibility of investigating LoRa-based off-grid messaging, but it also warns against overclaiming. LoRaResQ's distinct contribution is a bounded, alert-first workflow for a small rural community, not a new radio technology. Its next credible milestone is a measured three-node prototype whose interface, delivery states, relay policy, and limitations can be demonstrated transparently.

REFERENCES

- [1] F. Adelantado, X. Vilajosana, P. Tuset-Peiró, B. Martínez, J. Melià-Seguí, and T. Watteyne, "Understanding the limits of LoRaWAN," *IEEE Communications Magazine*, vol. 55, no. 9, pp. 34–40, 2017. doi: 10.1109/MCOM.2017.1600613.
- [2] A. Augustin, J. Yi, T. Clausen, and W. M. Townsley, "A study of LoRa: Long range & low power networks for the Internet of Things," *Sensors*, vol. 16, no. 9, Art. no. 1466, 2016. doi: 10.3390/s16091466.
- [3] Bluetooth SIG, "Bluetooth Core Specification, Version 5.4," technical specification, 2023. [Online]. Available: <https://www.bluetooth.com/specifications/specs/core-specification-5-4/>.
- [4] IEEE, "IEEE Standard for Information Technology–Telecommunications and Information Exchange Between Systems Local and Metropolitan Area Networks– Specific Requirements–Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications," IEEE Std. 802.11-2020, 2020. [Online]. Available: https://standards.ieee.org/standard/802_11-2020.html.
- [5] M. Jouhari, N. Saeed, M.-S. Alouini, and E. M. Amhoud, "A survey on scalable LoRaWAN for massive IoT: Recent advances, potentials, and challenges," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 3, pp. 1841–1876, 2023. doi: 10.1109/COMST.2023.3274934.
- [6] LoRa Alliance, "LoRaWAN specification," technical specification, 2020. [Online]. Available: <https://lora-alliance.org/wp-content/uploads/2020/11/lorawan1.0.3.pdf>.
- [7] M. Hammouti, O. Moussaoui, M. Hassine, and A. Betari, "Exploring open source and proprietary LoRa mesh technologies," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 34, no. 2, pp. 960–969, 2024. doi: 10.11591/ijeecs.v34.i2.pp960-969.
- [8] R. Pueyo Centelles *et al.*, "LoRaMoto: A communication system to provide safety awareness among civilians after an earthquake," *Future Generation Computer Systems*, 2021. doi: 10.1016/j.future.2020.07.040.
- [9] Meshtastic, "Meshtastic documentation: Mesh broadcast algorithm and configuration," online documentation, accessed Oct. 2, 2026. [Online]. Available: <https://meshtastic.org/docs/overview/mesh-algo/>.
- [10] R. Garzon Andosilla and J. de Jesús Rugeles Uribe, "A Meshtastic-based LoRa mesh system for smart campus applications: From solar-powered sensing to containerized data management," arXiv preprint arXiv:2605.20379, 2026. [Online]. Available: <https://arxiv.org/abs/2605.20379>.
- [11] Government of India, "Use of Low Power Equipment in the Frequency Band 865–868 MHz for Short Range Devices (Exemption from Licence) Rules, 2021," G.S.R. 853(E), 2021. [Online]. Available: <https://thc.nic.in/Central%20Governmental%20Rules/use%20of%20low%20power%20Equipment%20in%20the%20frequency%20band%20865%20to%20868%20MHz%20for%20Short%20Range%20Devices%20Exemption%20from%20Licence%20Rules,2021.pdf>.